



Diagnostic perspectives on mixed connective tissue disease with tuberculosis overlap

Shree V Dhotre, Pradnya S Dhotre, Ajay M Gavkare, Basavraj S Nagoba

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Shree V Dhotre, Department of Microbiology, Ashwini Rural Medical College, Hospital and Research Centre, Solapur 413006, Maharashtra, India

Pradnya S Dhotre, Department of Biochemistry, Ashwini Rural Medical College, Hospital and Research Centre, Solapur 413006, Maharashtra, India

Ajay M Gavkare, Department of Physiology, Government Medical College, Buldhana 443001, Maharashtra, India

Basavraj S Nagoba, Department of Microbiology, Maharashtra Institute of Medical Sciences and Research (Medical College), Latur 413531, Maharashtra, India

Co-first authors: Shree V Dhotre and Pradnya S Dhotre.

Corresponding author: Basavraj S Nagoba, PhD, Assistant Dean, Professor, Department of Microbiology, Maharashtra Institute of Medical Sciences and Research (Medical College), Vishwanathpuram, Ambajogai Road, Latur 413531, Maharashtra, India.

dr_bsnagoba@yahoo.com

Abstract

The recently published case report describing mixed connective tissue disease coexisting with tuberculosis (TB) provides an important contribution to the growing literature on complex autoimmune-infectious disease interactions. This letter expands on the diagnostic challenges highlighted by the authors by defining its added clinical value: Identifying practical diagnostic “red flags”, emphasizing parallel consideration of autoimmunity and infection, and proposing a pragmatic approach to evaluation in TB-endemic settings. Early immunological testing, differentiated pulmonary assessment, and multidisciplinary decision-making are essential when overlapping features obscure timely diagnosis and complicate therapeutic choices.

Key Words: Mixed connective tissue disease; Tuberculosis; Autoimmune disease; Immunosuppression; Diagnostic red flags

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Core Tip: This letter highlights practical diagnostic insights beyond the original case, including clinical red flags that should prompt reconsideration of a tuberculosis-only diagnosis, the value of early autoimmune serology, and the importance of parallel diagnostic pathways in tuberculosis-endemic regions.

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TO THE EDITOR

We read with great interest the article by Sial *et al*[1]. The authors are to be sincerely commended for presenting a well-documented, clinically rich, and carefully analyzed case that illustrates a highly challenging diagnostic scenario. Their effort to highlight the interplay between systemic autoimmunity and infectious disease is timely and of considerable clinical relevance, particularly in regions where tuberculosis remains highly prevalent.

The manuscript is written with clarity, supported by detailed laboratory profiles, and thoughtfully contextualized within the existing literature. The case is especially valuable because coexistence of mixed connective tissue disease (MCTD) with active tuberculosis (TB) is rare, and symptoms frequently overlap to such an extent that early diagnostic misdirection is almost inevitable. By sharing this case, the authors have contributed significantly to awareness regarding such atypical presentations and the pitfalls of anchoring bias in clinical decision-making.

Importantly, the detailed immunological workup presented, including high-titer ANA and anti-Sm/RNP positivity, provides a strong diagnostic foundation. The authors' integration of clinical, serological, and radiographic findings exemplifies the type of multidisciplinary approach needed for managing such complex cases.

The therapeutic pathway described; continuing anti-tuberculosis treatment alongside cautious corticosteroid use-reflects current best practices and demonstrates sound clinical judgment. I appreciate the authors' emphasis on patient education, treatment adherence, and long-term monitoring, all of which are crucial yet often omitted in case reports.

Beyond reinforcing these strengths, this letter seeks to add practical value by highlighting circumstances in which clinicians should actively reassess an initial TB-centered diagnosis and consider autoimmune disease in parallel, rather than sequentially.

CURRENT SITUATION AND CHALLENGES

As the authors rightly point out, pulmonary involvement in MCTD frequently presents with dyspnea, pleural effusion, interstitial lung disease, or alveolitis; features that closely resemble pulmonary TB. In high-burden regions, this overlap often results in delayed or missed diagnoses[2,3]. Several observational studies suggest that a substantial proportion of patients with MCTD undergo initial evaluation for infectious etiologies before autoimmunity is recognized[4], underscoring the risk of diagnostic anchoring in endemic settings.

The case also underscores a critical consideration: Immunosuppressive therapies, such as methotrexate, corticosteroids, and biologics, can increase susceptibility to TB reactivation[5,6]. When autoimmune symptoms overlap with infection, determining whether clinical deterioration reflects underlying disease progression, drug-induced immunosuppression, or an infectious process becomes exceptionally difficult.

Furthermore, constitutional symptoms such as fatigue, fever, weight loss; show substantial overlap between TB and autoimmune flares, heightening diagnostic uncertainty. The coexistence of hepatitis C in this patient further complicates the clinical picture, as chronic viral infections may both aggravate autoimmune manifestations and influence therapeutic decision-making.

In TB-endemic regions, lack of response to appropriate anti-tubercular therapy, atypical radiologic findings, or the presence of strong autoimmune serologic markers should prompt clinicians to pursue parallel diagnostic pathways rather than deferring autoimmune evaluation.

KEY ASPECTS OF DIAGNOSIS AND MANAGEMENT

Early comprehensive immunological workup

Prompt screening for ANA, extractable nuclear antigen panel, and disease-specific antibodies can prevent misdiagnosis and facilitate earlier identification of overlapping autoimmune disorders[3,7].

Awareness that dual pathology is not uncommon

TB may coexist with autoimmune rheumatic diseases, including MCTD, particularly in endemic settings, and should not preclude simultaneous evaluation for systemic autoimmunity when clinical features are discordant[8,9].

TB can mimic or exacerbate autoimmune disease

Granulomatous inflammation from TB may present with vasculitic rashes, anemia, or renal involvement; features similar to autoimmune flares[10].

Balanced approach to immunosuppression

The authors' management strategy; corticosteroids used concurrently with anti-tubercular therapy, reflects internationally recommended guidance[11].

Multidisciplinary collaboration is indispensable

In cases like this, rheumatology, infectious diseases, pulmonology, and hematology teams must work cohesively to avoid delays in diagnosis and inappropriate therapy.

CLINICAL AND THERAPEUTIC IMPLICATIONS

The case demonstrates the importance of: Avoiding premature anchoring on TB when radiological or respiratory symptoms appear; ordering autoimmune serology early when infection does not fully explain the clinical picture; recognizing that TB can precipitate autoimmune flares through immune dysregulation; ensuring close monitoring when corticosteroids are initiated in suspected or confirmed TB cases; providing structured follow-up and patient counselling, particularly in complex overlap syndromes; the authors' work enriches the literature and reinforces the need for improved diagnostic strategies in resource-limited, TB-endemic settings.

CONCLUSION

In addition to supporting the original case report, this letter emphasizes pragmatic diagnostic insights for clinicians practicing in TB-endemic regions, including recognition of red flags that warrant parallel evaluation for autoimmunity and infection. By clarifying these decision points, the discussion aims to reduce diagnostic delay and inappropriate therapeutic sequencing in patients with overlapping infectious and autoimmune features.

FOOTNOTES

Author contributions: Dhotre SV conceptualized and designed the study, developed the study outline, and coordinated manuscript preparation; Dhotre SV, Dhotre PS, and Nagoba BS made substantial contributions to the study design, data interpretation, and critical discussion of the manuscript; Dhotre SV and Nagoba BS drafted, critically revised, and edited the manuscript for important intellectual content and contributed to the literature review; all authors approved the final version of the manuscript to be published.

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Country of origin: India

ORCID number: Basavraj S Nagoba 0000-0001-5625-3777.

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